Name ____ Date ___________ Period _____

Parent Functions II: x^3 and $\sqrt[3]{x}$

TEACHER KEY | Algebra II | Unit 1 | Day 5 | TEK A2.2.A

Objective: I can graph the cubic and cube root parent functions from a table and analyze their key attributes, including their rotational symmetry about the origin.

Part 1 — The Cubic Function, $f(x) = x^3$

Fill in the table. Notice what a negative input produces.

x	-2	-1	0	1	2
$f(x) = x^3$	-8	-1	0	1	8

A negative input gives a **negative** output, because a negative cubed stays negative.

Part 2 — The Cube Root Function, $f(x) = \sqrt[3]{x}$

This undoes the cubic. Compare it to your Part 1 table.

x	-8	-1	0	1	8
$f(x) = \sqrt[3]{x}$	-2	-1	0	1	2

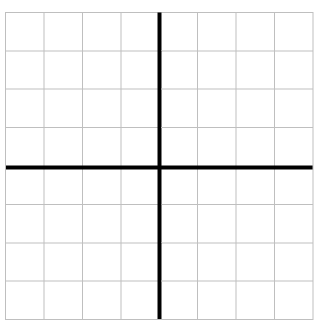
What do you notice next to Part 1? **the x and y values trade places**

We will come back to that on Day 10 - these two are **inverses**.

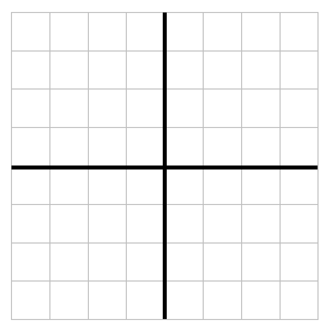
Part 3 — Graph Them Both

Each square is 1 unit. Both grids run -8 to 8 on each axis, so 1 square = 2 units.

A. $f(x) = x^3$ steep S-curve



B. $f(x) = \sqrt[3]{x}$ flatter S-curve



Passes through the **origin**

Mirror line between them: **y = x**

Part 4 — Compare the Key Attributes

Attribute	**$f(x) = x^3$**	**$f(x) = \sqrt[3]{x}$**
Domain (interval)	**$(-\infty, \infty)$**	**$(-\infty, \infty)$**
Range (interval)	**$(-\infty, \infty)$**	**$(-\infty, \infty)$**
x- and y-intercept	**(0, 0)**	**(0, 0)**
Symmetry	**rotational, 180°**	**rotational, 180°**
Maximum or minimum?	**neither**	**neither**

Increasing / decreasing	**always increasing**	**always increasing**

These two share every attribute above. The difference is the **steepness** of the curve.

Part 5 — Compare to Day 4

1. Why does $\sqrt[3]{x}$ accept negatives when $\sqrt{x}$ does not? **a negative cubed is negative, but no real number squared is negative**

2. Which of the four functions so far have a minimum? **$\sqrt{x}$ and $|x|$**

Practice

Evaluate and explain. Show your work where there is room.

1. Evaluate: $(-3)^3$ = **-27** $\sqrt[3]{-27}$ = **-3** $\sqrt[3]{64}$ = **4** $\sqrt[3]{0}$ = **0**

2. Solve $x^3 = 125$. x = **5** How many real answers? **1**

3. Solve $x^2 = 25$. x = **5 and -5** How many real answers? **2**

4. Items 2 and 3 give different counts. Explain why cubing behaves differently from squaring.

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5. $f(x) = x^3$ on the interval [-1, 3]: minimum **-1** maximum **27**

6. True or false: every real number has exactly one real cube root. **true**

7. Which of the four parent functions so far accept negative inputs?
$|x|$, x^3, and $\sqrt[3]{x}$ (not $\sqrt{x}$)

8. Solve $x^3 = -8$. x = **-2** How many real answers? **1**

9. Evaluate: $\sqrt[3]{-1000}$ = **-10**

10. A cube-shaped storage box has volume $V = x^3$. If V = 343 in^3, what is the side length x? **7 in**

11. True or false: a negative number has exactly one real cube root, but no real square root. **true**

Exit Ticket

A function has domain and range both $(-\infty, \infty)$, passes through the origin, has 180° rotational symmetry, and no maximum or minimum.

Name two parent functions this could be: **x^3 and $\sqrt[3]{x}$**

What extra information would tell them apart? **a point such as (2, 8) versus (8, 2)**